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Assignment II

(4)

1. **Visualizing Amounts:**

Aim: To visualize and compare quantities using horizontal bar charts, grouped bar charts and stacked bar charts

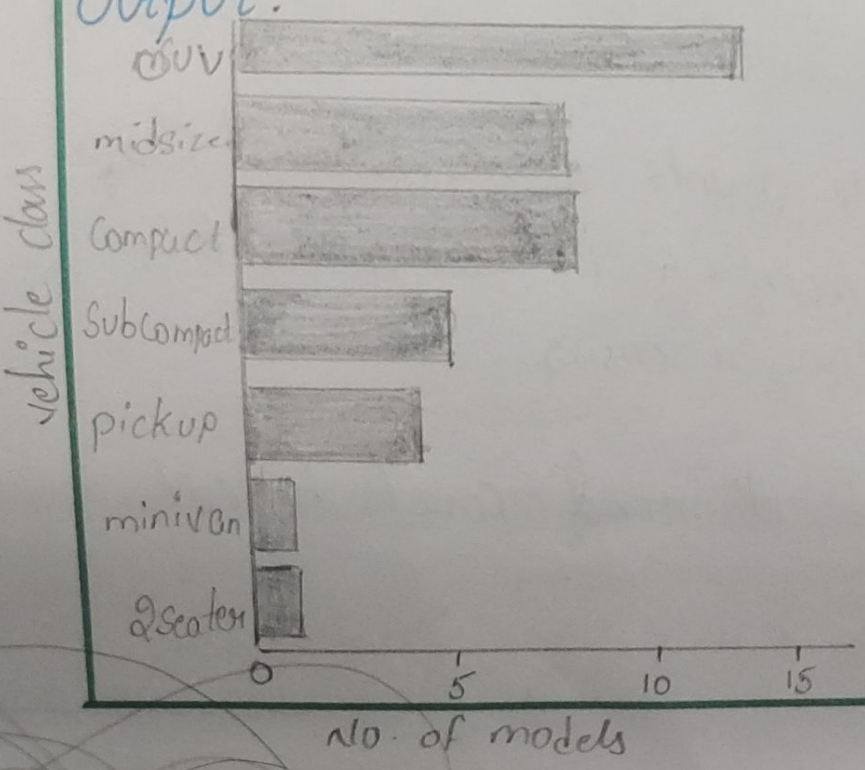
(a) Using mpg.draw a horizontal bar chart of the number of models in each vehicle class. sorted from most to least frequent

Program:

```
data(mpg)
model_count <- mpg %>%
  distinct(manufacturer, model, class) %>%
  count(class, name = "total_models") %>%
  arrange(desc(total_models))

ggplot(model_count,
       aes(x = fct_reorder(class, total_models), y = total_models)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Number of vechile Model in each class", x = "Vehicle class",
       y = "Number of models") +
  theme_minimal()
```

Output:



Interpretation:

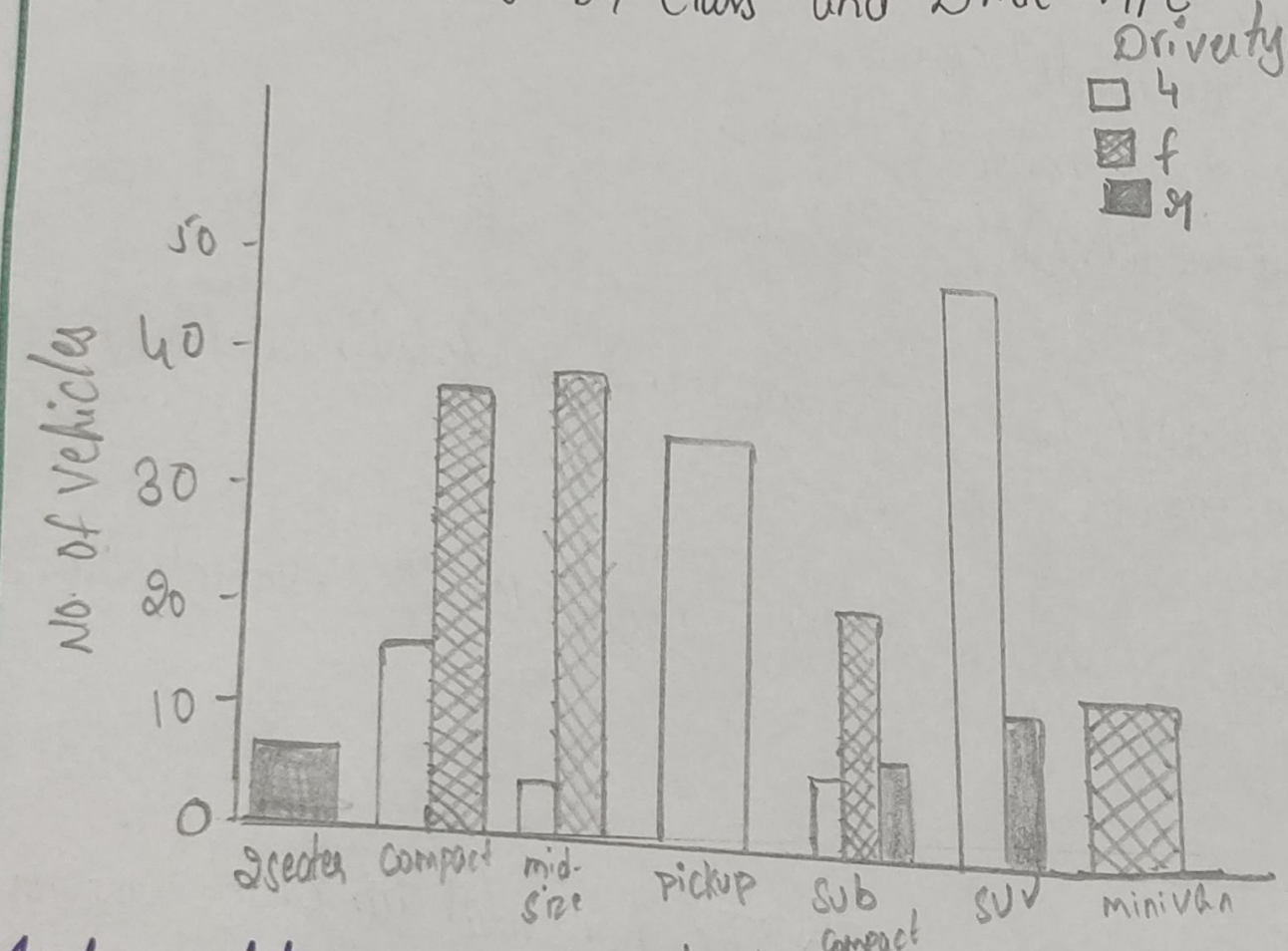
* The horizontal bar chart compares the no. of distinct vehicles models accross different vehicle classes
* SUV has highest no. of models

6. Create a grouped bar chart showing the count of vehicles in each class broken down by drive type (drv)

Program:

```
ggplot(mpg, aes(x= class, fill = drv))+
  geom_bar(position, "dodge")+
  labs(title: "Vehicle by class and drive type", x = "Vehicle class",
    Y= "Number of vehicles", fill = "Drive Type")+
theme_minimal()
```

Output:



Interpretation:

The grouped chart compares vehicle counts accross different classes and drivetypes. Different represent front-wheel, rear wheel, four wheel drive vehicles, making Comparison easier

C. Create a stacked bar chart of diamond counts by cut. Stacked by clarity

Program

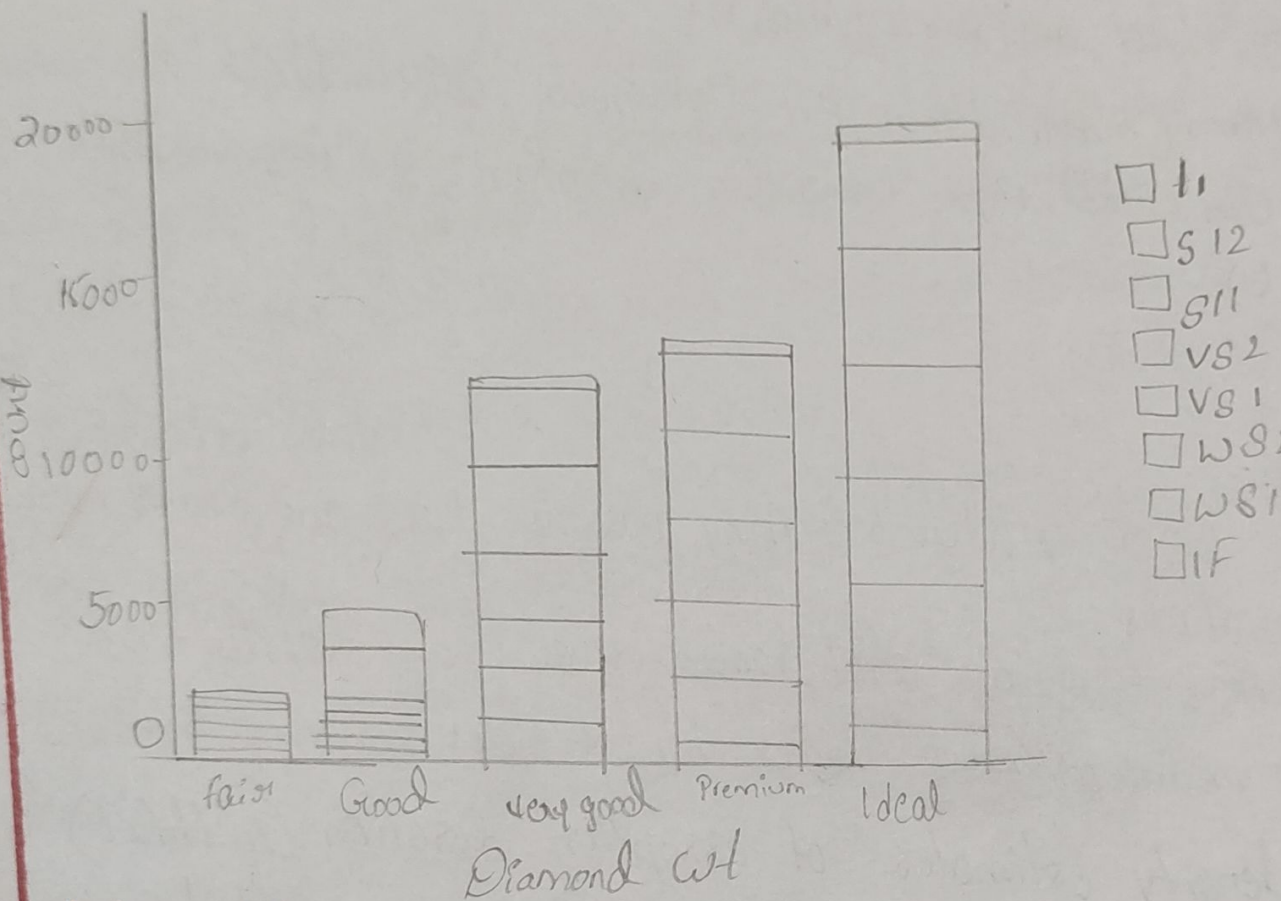
```
data(diamonds)
ggplot(diamonds
  geom_bar()+
  labs(title =
    Y="Count"
theme_minim
```

Output

Program

```
data(diamonds)
ggplot(diamonds, aes(x=cut, fill=clarity))+
  geom_bar()+
  labs(title="Diamond counts by cut and clarity", x="Diamond cut",
  y="count", fill="clarity")+
theme_minimal()
```

Output:



Interpretation:

The Stacked bar chart displays the total no. of diamonds for each cut category. Each bar divided into different clarity grades, showing how clarity contributions to the total diamont count

2. Visualizing a Single Distribution:

a. Plot histogram of eruption duration using three bin widths (0.25, 0.05, 1). Discuss the effect of bin width.

Program:

```
data(faithful)
P1 <- ggplot(faithful, aes(x=eruption))+
      geom_histogram(binwidth=0.05, fill="sky blue", colour="black")+
      labs(title="bin width=0.05", x="encryption duration", y="frequency")+
theme_minimal()

P2 <- ggplot(faithful, aes(x=eruptions))+
   geom_histogram(binwidth=0.25, fill="orange", color="black")+
   labs(title="Binwidth=0.25", x="Eruption duration", y="frequency")+
 theme_minimal()
P3 <- ggplot(faithful, aes(x=eruptions))+
   geom_histogram(binwidth=1, fill="lightgreen", color="black")+
   labs(title="Bin Width=1", x="Eruption duration", y="frequency")+
theme_minimal()
P1=P2+P3
```

Interpretation:

* Bin width 0.05 produces many narrow bars and shows more detailed variations
* Bin width 0.25 provides a balanced view of distribution
* Bin width 1 produces fewer wider bars & hides some details

(b) Plot kernel density estimates of eruption duration (faithful) with three bandwidth adjustments (0.25, 1, 3) on one plot

Aim: To understand the distribution of a single numerical variable using histogram and Kernel density estimation

Program:

```
ggplot(faithful, aes(x=eruption))+
   geom_density(adjust=0.25, colour="red", linewidth=1)+
```

```
geom_density(adjust=1, color="black", linewidth=1)+
geom_density(adjust=3, color="darkgreen", linewidth=1)+
labs(title="Density Estimation with Different Bandwidths",
     x="Eruption Duration",
     y="Density")+

theme_minimal()
annotate("text", x=2, y=0.9,
         label="red:0.25 | blue:1 | Green:3")
```

Interpretation:
Bandwidth adjustment controls the smoothness of the density curve. An adjustment of 0.25 produces a more detailed and irregular curve, while 1 gives a balanced distribution. An adjustment of 3 produces a smoother curve but may hide important distribution patterns

3. Visualizing Multiple Distributions

c **Aim:** To compare multiple data distributions using density plots, barplots and violin plots

a. Visualize multiple distributions simultaneously: draw overlapping density plots of Sepal length for three iris species using partial transpalency

Program:

```
data(iris)
ggplot(iris, aes(x=Sepal.Length, fill=Species, color=Species))+
geom_density(alpha=0.4,
             linewidth=1)+
labs(title="Sepal Length Distribution by Iris Species",
```

```
    x = "Sepal length",
    y = "Density",
    fill = "Species",
    color = "Species") +
  theme_minimal()
```

Interpretation:
The overlapping density plots compare Sepal length distributions of setosa, versicolor and virginica. Transparency helps identify overlaping regions and diff in the distribution of each species

b Using ToothGrowth, draw boxplots of tooth length by dose, split by supplement type with the raw data points jittered on top

Code:

```
ggplot(ToothGrowth, aes(x = dose, y = len, fill = supp)) +
  geom_boxplot(position = position_dodge(0.8), width = 0.6) +
  geom_jitter(aes(color = supp),
              position = position_jitterdodge(
                jitter.width = 0.15,
                dodge.width = 0.8),
              alpha = 0.6) +
  labs(title = "Tooth length by Dose and supplement", x = "Dose",
       y = "Tooth length", fill = "Supplement", color = "Supplement") +
  theme_minimal()
```

Interpretations:
The boxplots compare tooth length at different doses for 2 supplements OJ & VC. Jittered points represent individual observation and help identify the spread of raw data. The median and variability can be compared b/w supplement groups

Draw violin plots with a narrow boxpl marked as a red

Code:

ggplot(Toot
ae

gg

Draw violin plots of tooth length by dose (ToothGrowth) with a narrow boxplots inside each violin and the mean marked as a red point

(4)

Code:

```
ggplot(ToothGrowth,
       aes(x=factor(dose),
           y=len))+
  geom_violin(fill="lightblue",
              color="black")+
  geom_violin(width=0.12,
              fill="white")+
  stat_summary(fun.mean,
               geom="point",
               color="red",
               size=3)+
  labs(title="Tooth length Distribution by Dose",
       x="Dose",
       y="Tooth length")+
  theme_minimal()
```

Interpretation:

The violin plots shows the distribution and density of toot length at different doses. The narrow boxplots displays the median and quartiles while red points indicates the mean tooth len. This helps compare the distribution and avg vals accross doses

4. Visualizing Proportions

Aim: To represent proportions and percentage distribution using pie charts, 100% stacked bar charts and stacked density plots

a. Using mpg draw pie chart of the share of vehicles by drive type (drv) with percentage labels on each size.

Code:

```
driven_data <- mpg %>%
  count(drv) %>%
  mutate(percentage = n/sum(n) * 100, label = paste0(round(percent-
      -age, 1), "%"))
ggplot(drive_data, aes(x = "", y = n, fill = drv)) +
  geom_col(color = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = label), position = position_stack(vjust = 0.5),
  color = "white", size = 5) +
  labs(title = "percentage of vehicle by drive type", fill = "Drive
       Type") +
theme_void()
```

Interpretation:

The pie chart represents the percentage share of vehicles belongs to each drive type. Each slice shows the proportion of front wheel, rear-wheel & 4 wheel drive vehicles in dataset

b. Draw a 100%. stacked bar plot showing the proportion of each clarity grade within every cut of diamonds

Code:

```
ggplot(diamonds, aes(x=cut, fill=clarity))+
  geom_bar(position="fill")+
  scale_y_continous(labels=scales::percent)+
  labs(title="clarity Proposition within each diamond cut",
       x="Diamond cut",
       y="Percentage",
       fill="clarity")+
  theme_minimal()
```

Intespretation:

The 100% stacked bar chart display the proposition of each clarity grade within every diamond cut category. Every bar has the same total height of 100% allowing easy comparison of clarity distributions accross different cuts

C. Draw a stacked density plot showing how the proposition of each cut changes along carat

Code:

```
ggplot(diamonds, aes(x=carat, fill=cut))+
  geom_density(position="fill", alpha=0.8)+
  Scale_y_continuous(labels=scales::percent)+
  labs(title="proportion of Diamond cuts along carat",
       x="carat", y="proportion", fill="Diamond cut")+
  theme_minimal()
```

Intespretation:

The stacked density plot show how the relative proportions of different diamond cut categories vary with carat weight. The vertical axis represents propotions the combinsity at each carat

5. Nested Proportions and X-Y Relationships

a. Visualize nested proportions of vehicles by class and manufa... (mpg) using a treemap

Code:

```
treemap_data <- mpg %>%
ggplot(tree_map_data, aes(area = total, fill = class, subgroup = manufa-
        -cturer, label = manufacturer)) +
geom_treemap_subgroup_border(color = "white", size = 2) +
```

Interpretation: The treemap represents the histohierarchial distribution of vehicles by class manufacturer. The size of each rectangle represents the no. of vehicles while diff colours identify vehicles classes

b. Using mpg, draw a scatterplot of engine displacement vs highway mileage colored by deive type and add linear regression line for each group

Code:

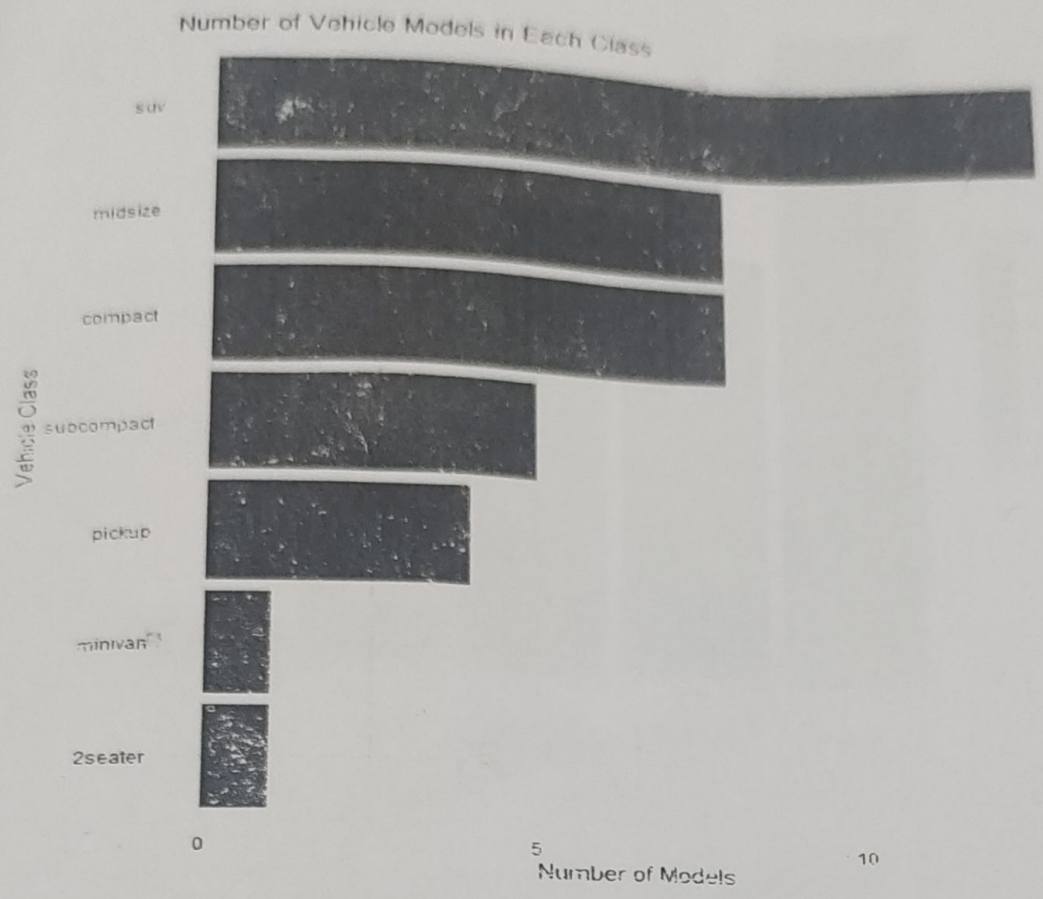
```
ggplot(mpg, aes(x = displ, y = hwy, color = drv)) +
  geom_point(size = 2, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  labs(title = "Engine Displacement vs Highway Mileage",
       x = "Engine Displacement", y = "Highway Mileage (MPG)",
       color = "Drive Type") +
theme_minimal()
```

Interpretation:

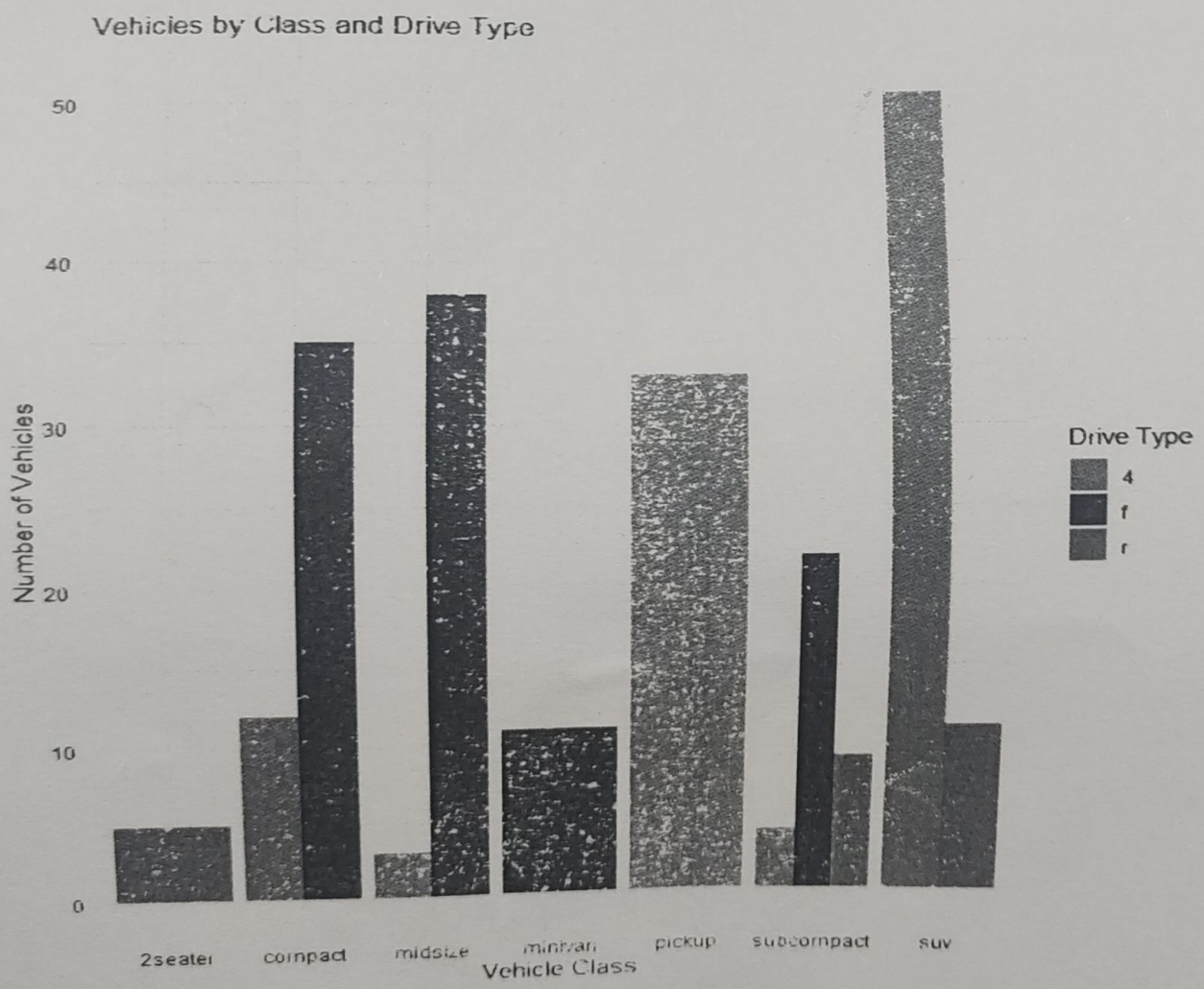
The Scatterplot shows the relationship blw engine displacement & highway fuel effiency for different drive types. The regression line indicate the overall trend for each drive type.

Q1. Visualizing Amounts

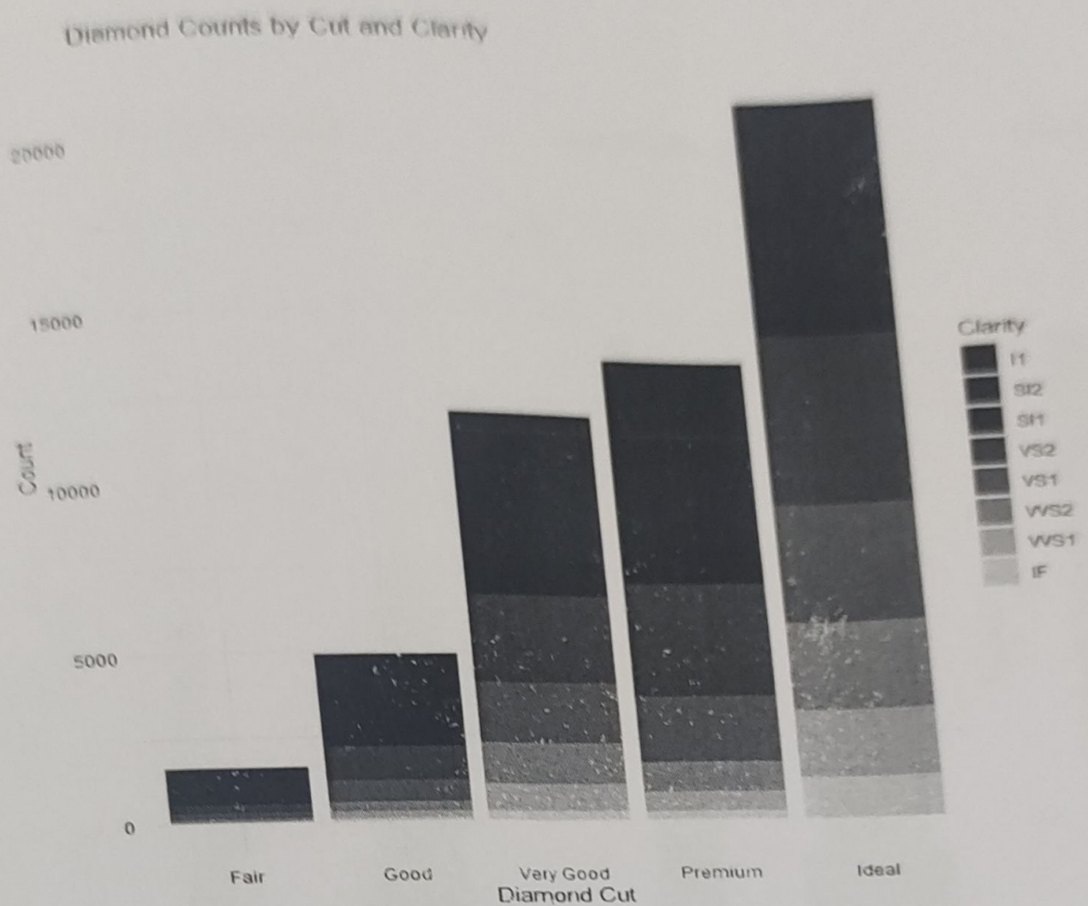
(a)



(b)

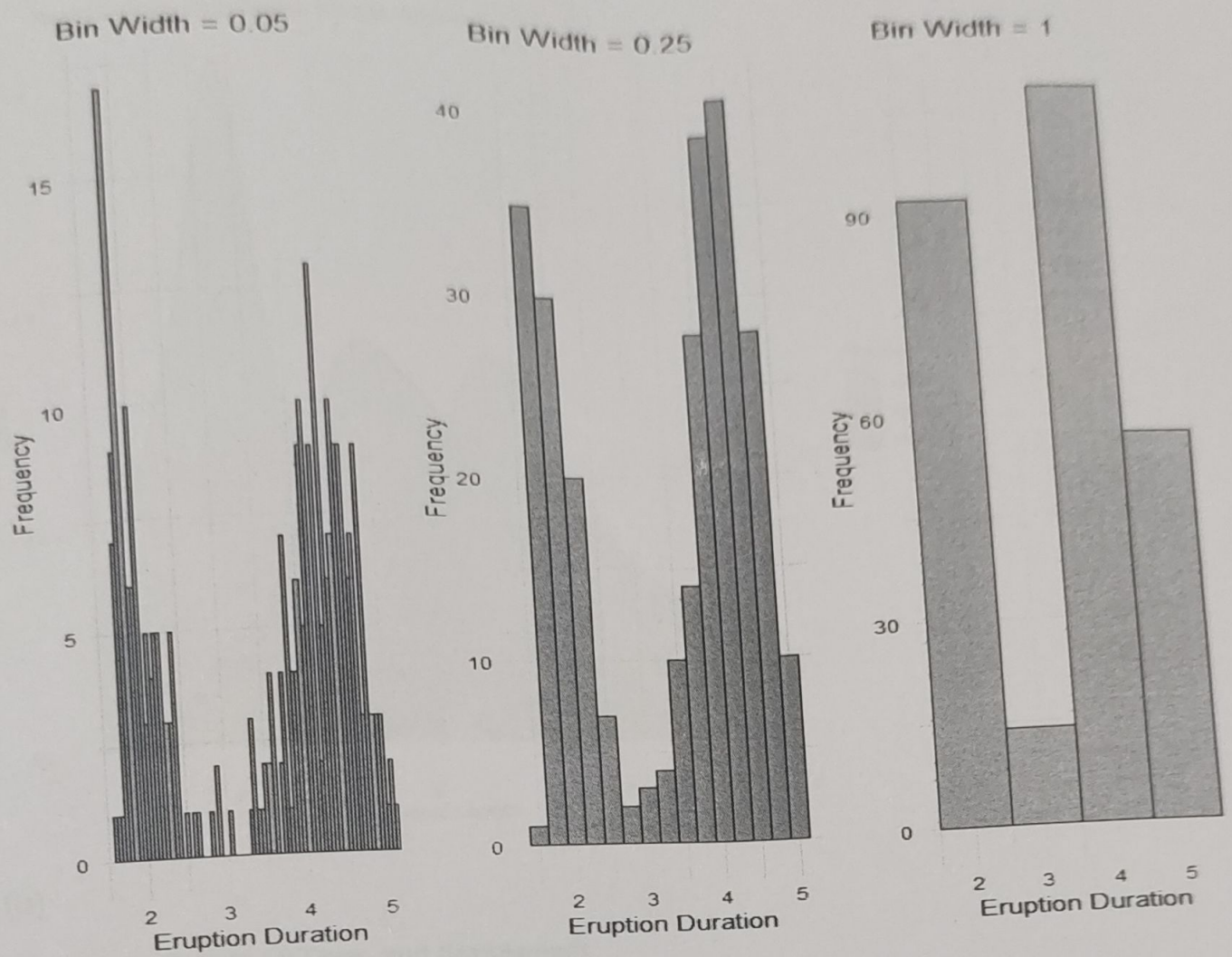


(c)

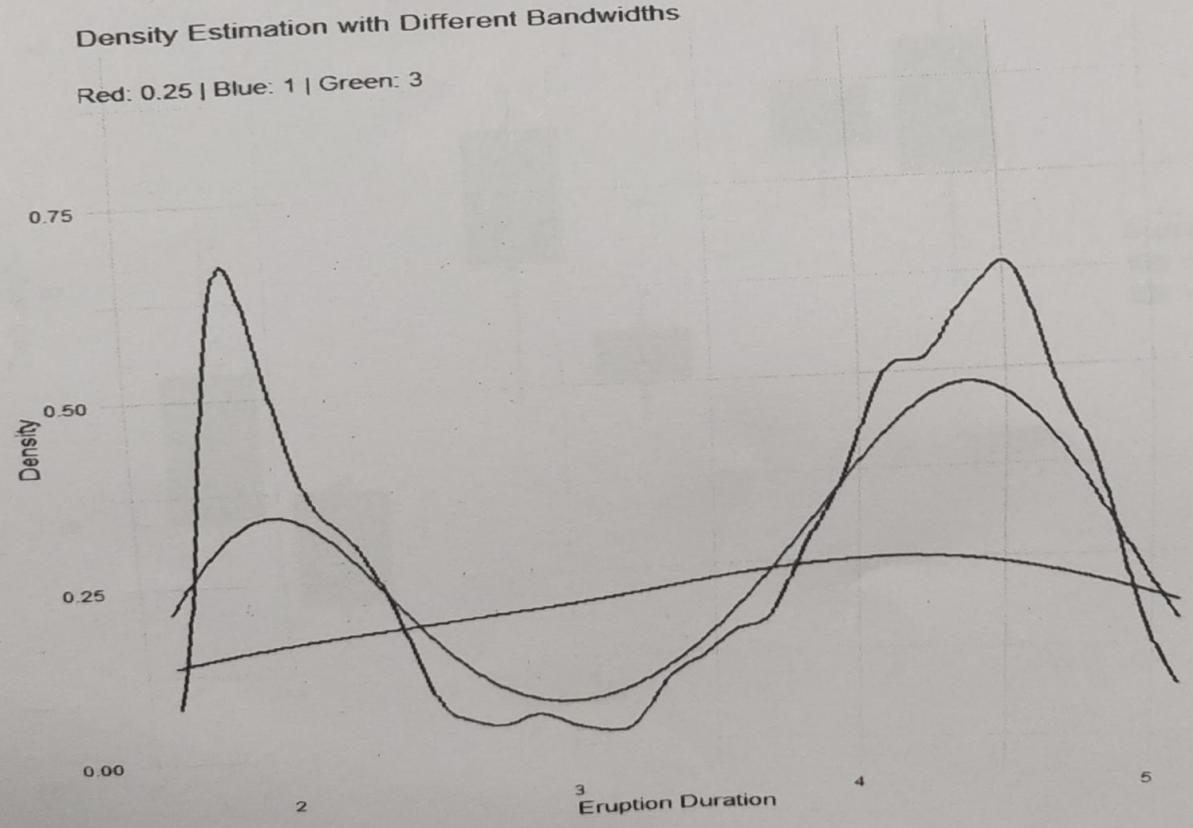

Diamond Counts by Cut and Clarity
20000
15000
10000
5000
0
Count
Fair
Good
Very Good
Premium
Ideal
Diamond Cut
Clarity
I1
SI2
SI1
VS2
VS1
VVS2
VVS1
IF

Q2. Visualizing a Single Distribution

(a)

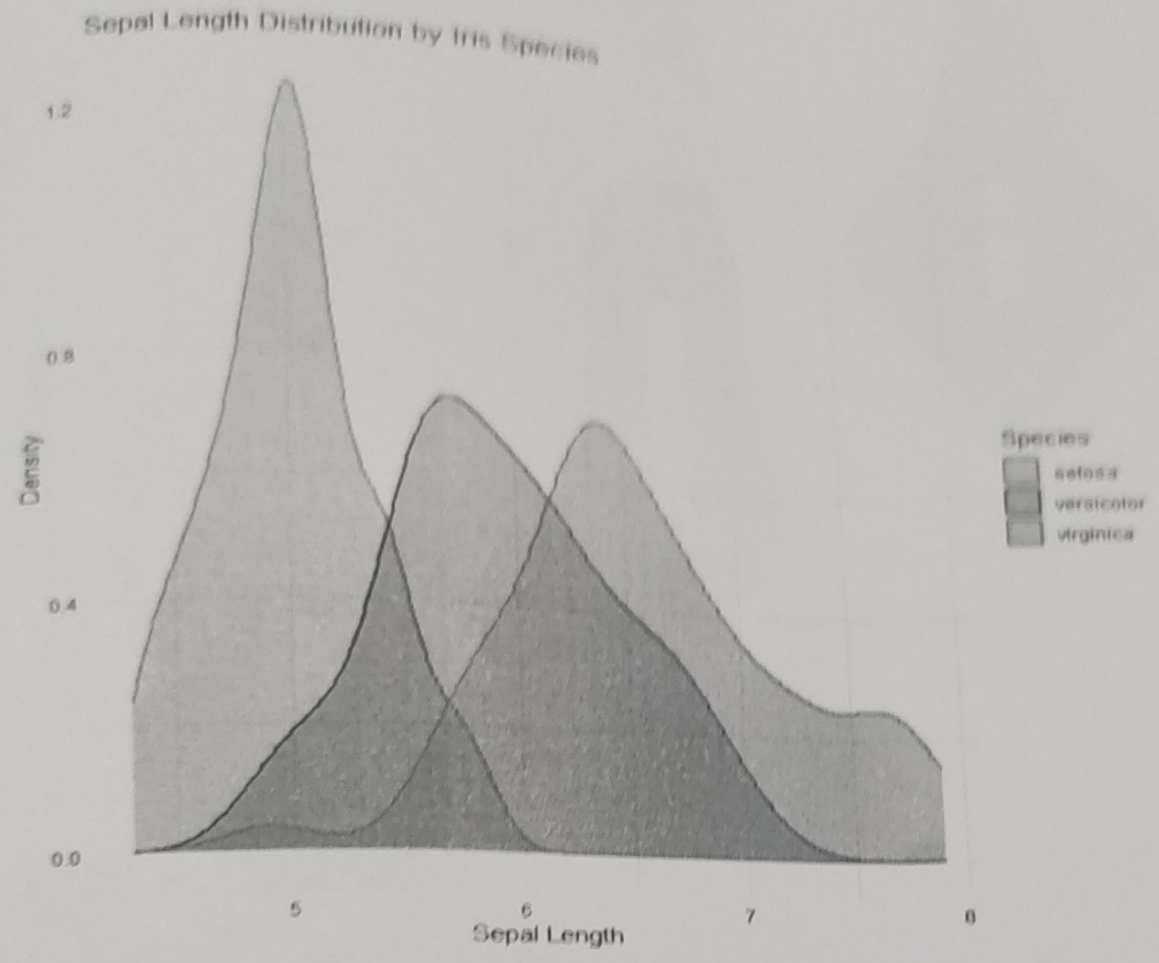


(b)

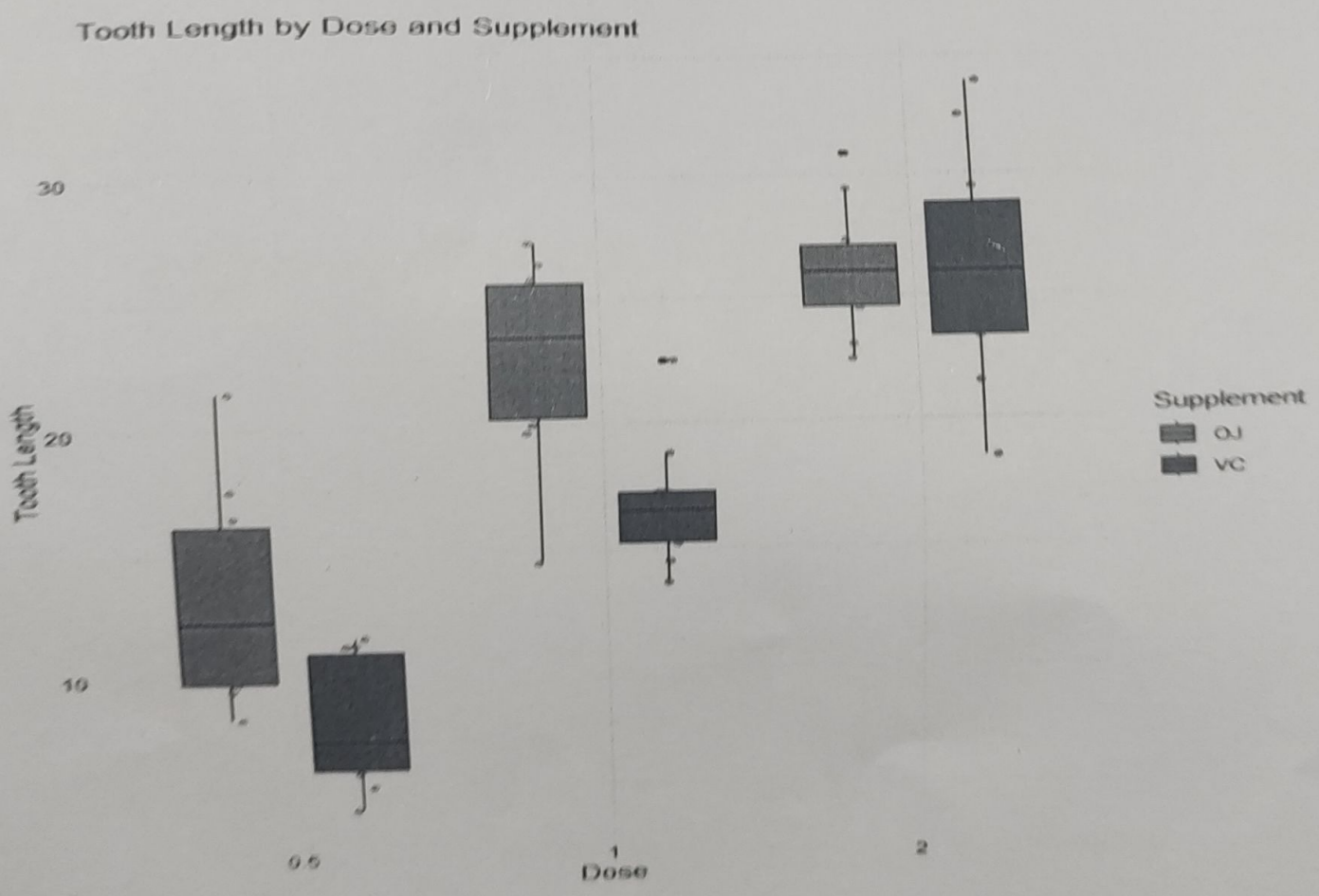


Q3. Visualizing Multiple Distributions

(a)

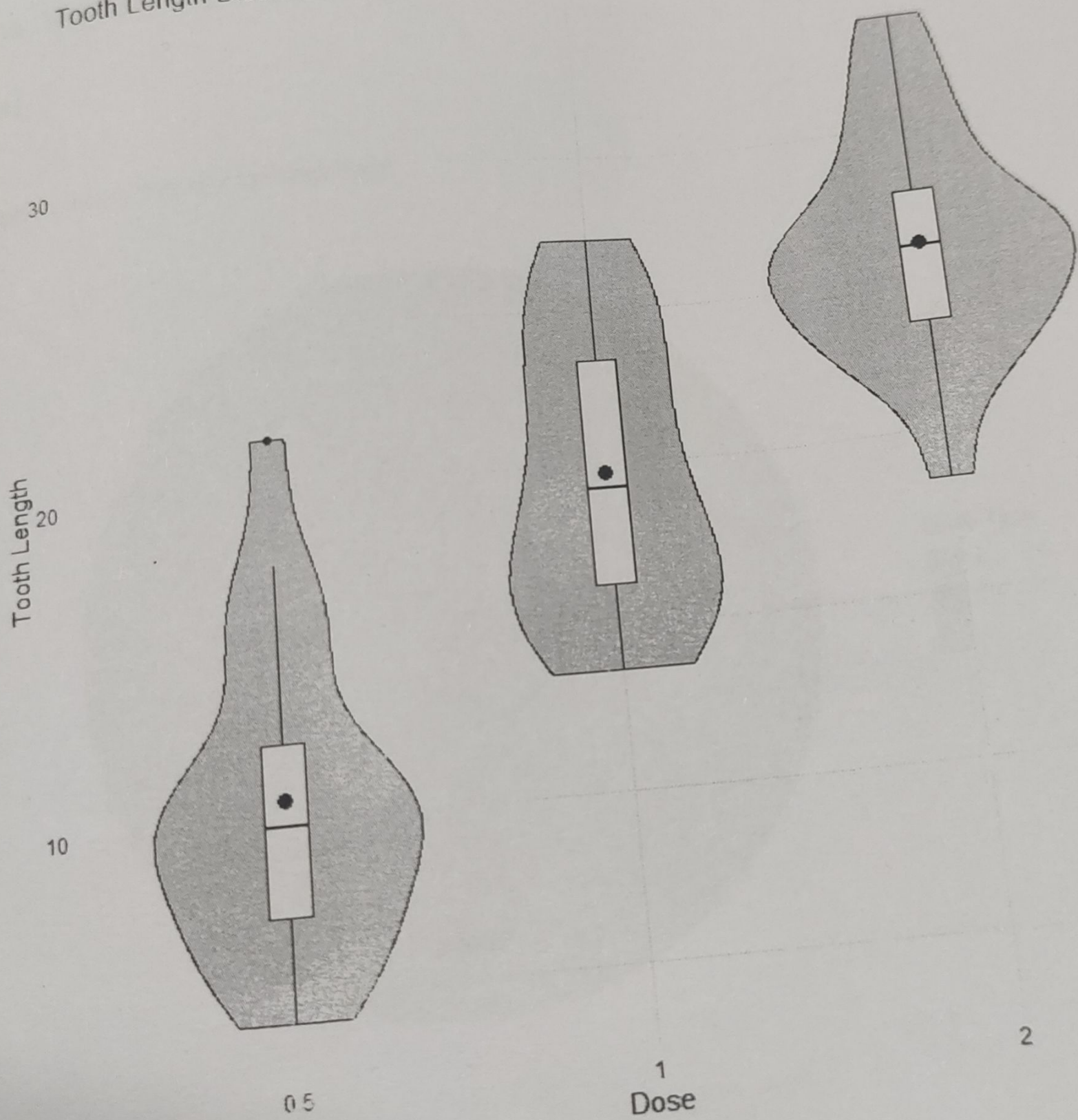


(b)

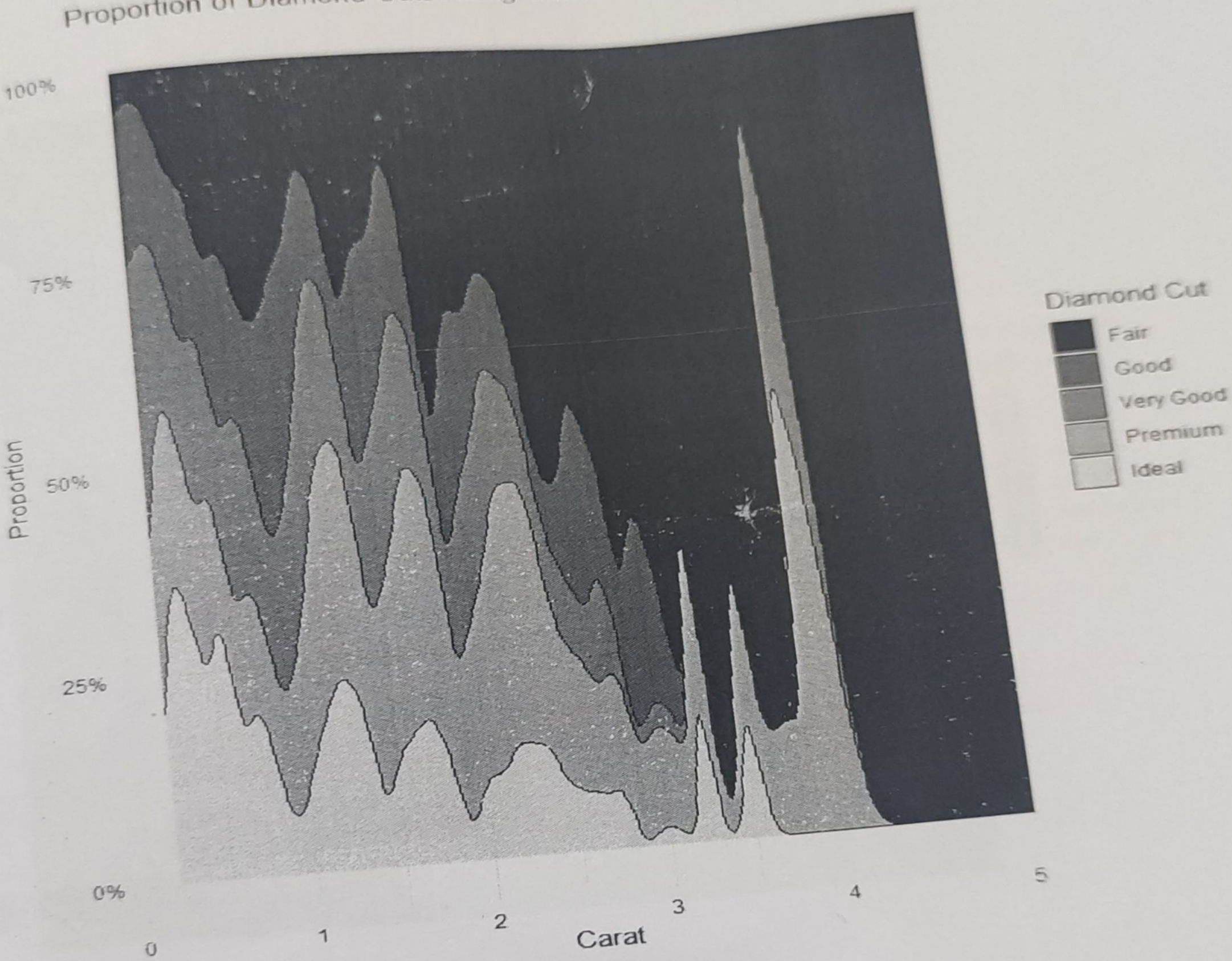


(c)

Tooth Length Distribution by Dose

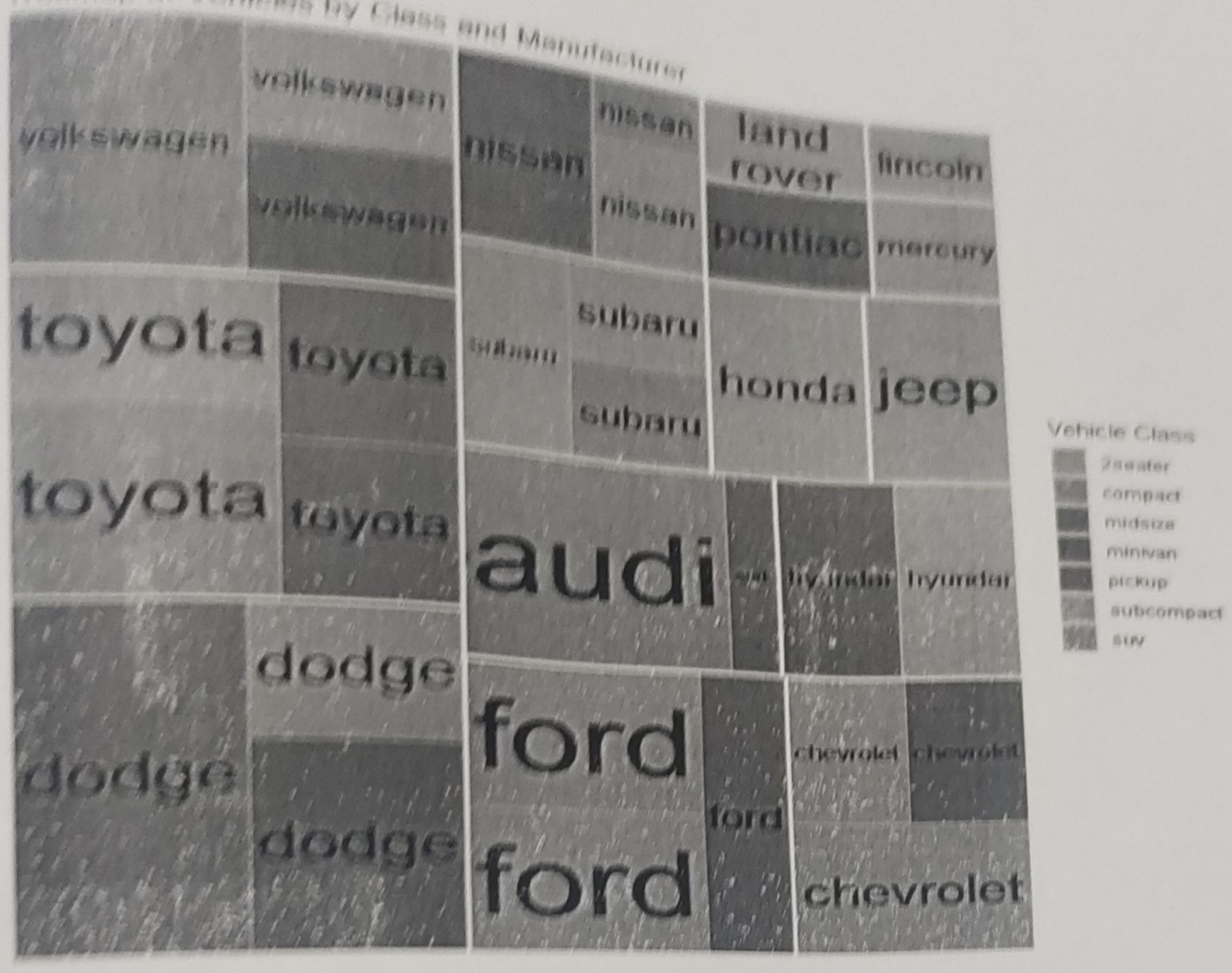


(c)


Proportion of Diamond Cuts Along Carat
100%
75%
50%
25%
0%
Proportion
0
1
2
3
4
5
Carat
Diamond Cut
Fair
Good
Very Good
Premium
Ideal

Q5. Nested Proportions and X–Y Relationships

(a)



(b)

